RESQ

Connected Communities: A Unified Disaster Resilience & Response System

Complete Feature Specification

This document consolidates the features described in the SANKALP project PDF for the RESQ disaster resilience platform.

1. Home Dashboard

- Emergency SOS button
- Current disaster/risk status
- Real-time alerts
- Weather information
- Quick access to emergency services
- Nearby shelters and hospitals
- Current user location
- Recent incidents
- Quick access to important services

2. AI-Driven Risk Assessment

- Localized flood risk prediction
- Localized landslide risk prediction
- Heatwave risk assessment
- Rainfall analysis
- Terrain/elevation analysis
- Population data consideration
- Historical disaster data
- Localized and explainable risk score
- Risk explanation for the user

3. Real-Time & Predictive Monitoring

- Weather data monitoring
- IoT sensor streams
- Soil-moisture monitoring
- Terrain/tilt sensor monitoring for landslide-prone zones
- Real-time flood, heatwave and landslide risk monitoring
- Predictive risk updates

4. Interactive Live Risk Map

- Current risk zones
- Flood-risk areas
- Landslide-risk areas
- Emergency shelters
- Hospitals
- Blood banks
- Blocked roads
- Reported hazards
- Lower-risk/safe routes
- User location
- All important disaster information in one map view

5. Incident & Hazard Reporting

- Report floods, landslides and other hazards
- Report road blockages and damage
- Upload/take incident photographs
- Attach GPS location
- Add incident description
- Submit citizen reports
- AI-assisted verification
- Authority validation
- Incident prioritization
- Track report from submission to resolution

6. SOS & Emergency Help Tracking

- SOS request
- Emergency location sharing
- Route SOS/requests to relevant organizations
- Emergency assistance workflow
- Response status tracking
- Continuous emergency updates

7. Safe Places

- Find nearby shelters
- Find nearby hospitals
- Find blood banks
- Show safe locations
- Show distance/location information
- Provide directions where available

8. Lower-Risk Route

- Identify high-risk areas
- Consider blocked roads
- Consider reported hazards
- Suggest safer/lower-risk routes
- Display safer routes on the map

9. Citizen–Authority Bridge

- Connect citizens with authorities
- Photo + location based reporting
- Verification and prioritization
- Response coordination
- Resolution tracking

10. Multi-Organization Network

- Role-based dashboards
- Authorities
- Hospitals
- NGOs
- Suppliers
- Share verified information
- Coordinate emergency response
- Match community needs with available resources
- Track response progress

11. Resilient Alert & Notification System

- In-app emergency notifications
- Push alerts
- Localized disaster warnings
- Risk-based alerts
- Multilingual alerts
- SMS fallback when mobile internet is unavailable but cellular service remains

12. Offline-Ready Communication

- Offline capabilities for emergency coordination
- Maintain access to important emergency information
- Support actions when local internet networks fail
- Queue/synchronize information when connectivity returns
- SMS fallback where cellular service is available

13. Multilingual Access

- English
- Hindi
- Marathi
- Multilingual alerts and guidance

14. Family Safety Group

- Create a family/group
- Add members
- Stay connected with loved ones
- Share safety/location information
- Emergency notifications for group members

15. Climate → Health Risk

- Flood → waterborne disease risk
- Heatwave → heat-stress risk
- Preventive risk indicators
- Information is for prevention and awareness, not medical diagnosis

16. Damage & Resource Intelligence

- Imagery-based damage assessment
- Flood/damage assessment
- Data-assisted shelter deployment
- Data-assisted medical resource deployment
- Data-assisted supply deployment

17. Actionable Disaster Guidance

- Pre-disaster guidance
- During-disaster guidance
- Post-disaster guidance
- Structured, clear safety instructions
- Actionable precautions

18. Closed-Loop Disaster Lifecycle

- PREDICT
- EXPLAIN
- ALERT
- REPORT
- RESPOND

- RECOVER
- Track emergency progress through the lifecycle

Core RESQ Workflow

Stage	Main Function
1. Collect & Analyse	Official alerts, weather/climate data, citizen reports and organization updates.
2. Assess & Inform	Identify local risks, generate risk maps, provide alerts, precautions and safer routes.
3. Connect & Coordinate	Route SOS/requests, match needs with organizations/resources and track response.
4. Support & Respond	Share verified information, assist shelters/hospitals/supplies and support recovery.

Key Innovation Areas

- AI multi-factor risk intelligence
- Real-time + predictive monitoring
- Interactive risk map
- Citizen–authority bridge
- Multi-organization coordination
- Resilient alerts with SMS fallback
- Climate-to-health risk awareness
- Damage and resource intelligence
- Closed-loop disaster resilience

Sample User Workflow

- Home Dashboard → Report Incident → Risk Alert → Live Risk Map → Safe Places → Family Group

Project Goal

Turn real-time information into real-world action and make communities more resilient to climate and disaster risks by connecting citizens with authorities, hospitals, NGOs and suppliers.